

Motivation

Global Prevalence rate (number of people living with cirrhosis per 100,000 population)

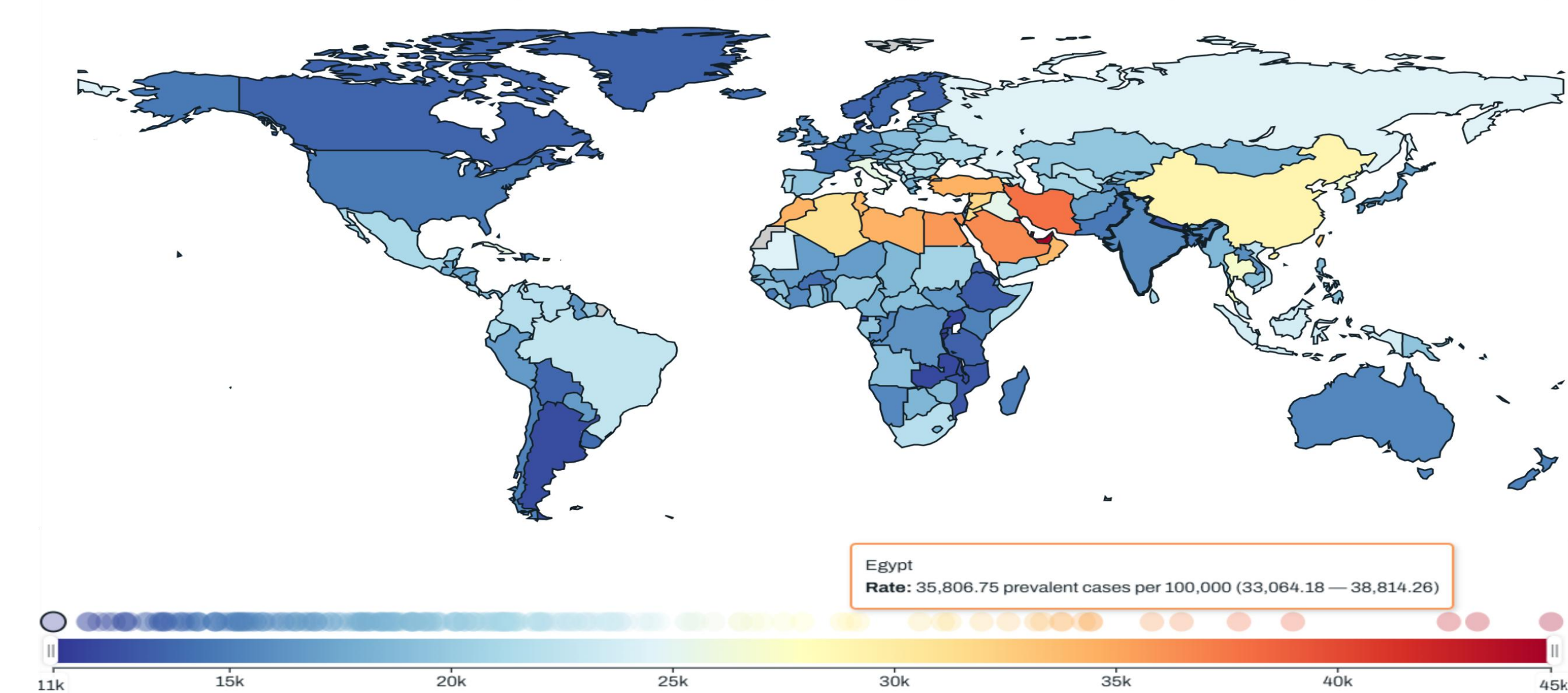


Figure 1: Global prevalence of cirrhosis and chronic liver disease, 2023. This map depicts the estimated number of people living with cirrhosis per 100,000 population worldwide. High-burden regions including North Africa, the Middle East, South Asia, and Central Asia reflect the ongoing impact of chronic viral hepatitis, alcohol use, and metabolic liver disease. Lower prevalence in North America, Western Europe, and East Asia highlights the benefits of vaccination and early screening. **Data source:** IHME Global Burden of Disease (GBD 2023).

Contribution

This research contributes to the development of **3UResNet**, an advanced ensemble deep learning model designed for multiclass segmentation of cirrhosis on 2D T2-weighted MRI scans.

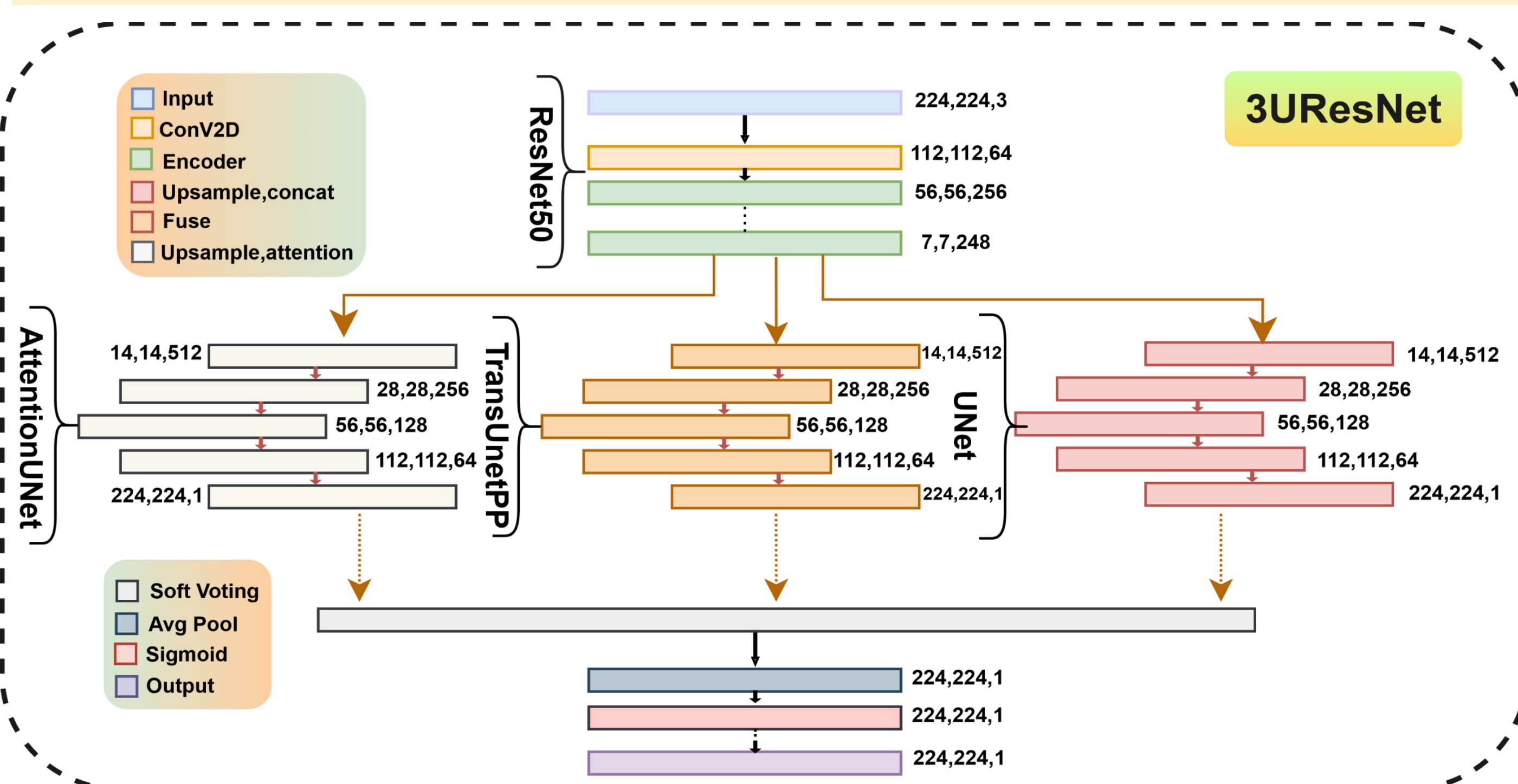
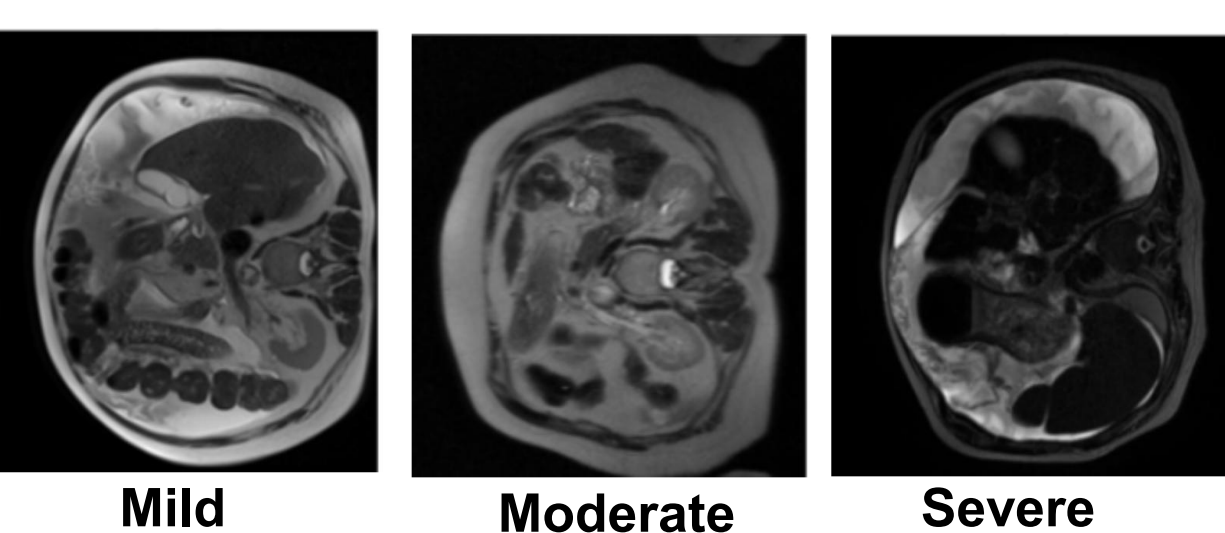


Figure 2: The proposed 3UResNet architecture. The model integrates three hierarchical U-Net streams—U-Net, TransUNet++, and Attention U-Net—combined through soft voting to enhance boundary precision and feature diversity. Each branch utilizes upsampling with concatenation and attention mechanisms to integrate multi-resolution features. The outputs are combined via soft voting, average pooling, and sigmoid activation to produce the final segmentation output. Each encoder-decoder block employs convolution, upsampling, and attention mechanisms to preserve contextual details while refining lesion boundaries. This multi-stream design enables robust feature learning across varying cirrhosis severity levels, improving accuracy, stability, and interpretability in medical image segmentation.

Data Descriptions



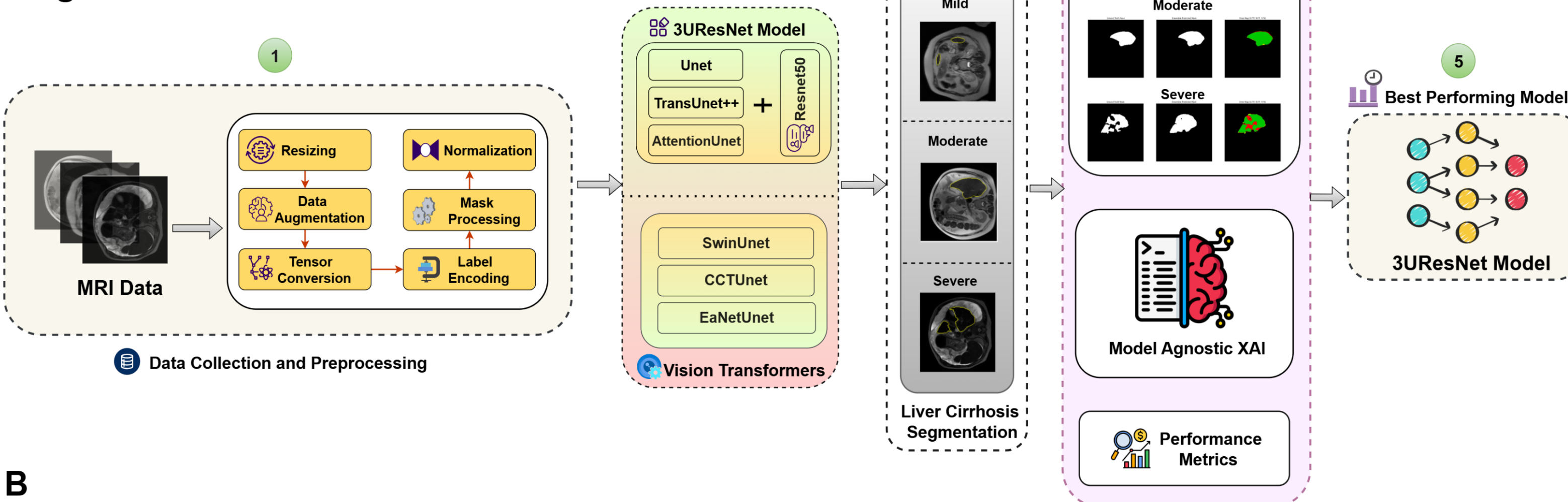
- 318 T2W MRI scans from 339 patients
- Includes 5,364 training samples
- 674 validation samples
- 664 test samples

CirrMRI600+

Publicly Available at: <https://osf.io/cuk24>

Methodology

A Segmentation Block



B Classification Block

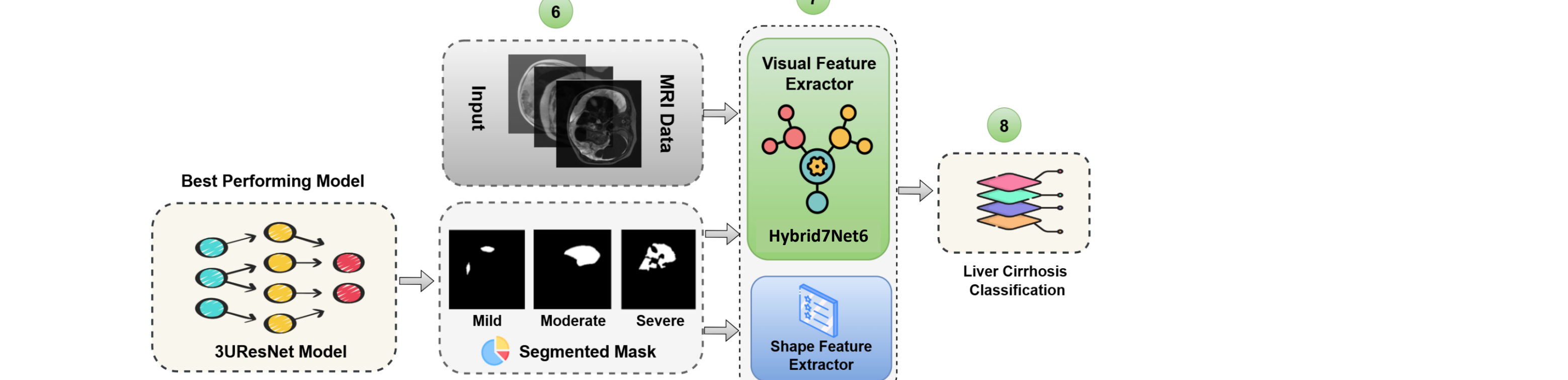


Figure 3: Deep learning pipeline for liver-cirrhosis segmentation and severity classification from MRI data. The workflow begins with MRI preprocessing (resizing, normalization, and label encoding) to create standardized input images and masks. A hybrid 3UResNet-Vision Transformer framework performs liver segmentation by combining U-Net, TransUNet++, and Attention U-Net architectures to enhance spatial precision and contextual learning. The output masks are categorized as mild, moderate, or severe cirrhosis and validated through explainable AI (XAI)-based error analysis and performance metrics. Segmented regions are passed to a Hybrid7Net6-based feature extractor, where both texture and shape descriptors are learned. Finally, a Feature Fusion Classifier (FFC) integrates multimodal features to produce the final disease-severity prediction. This end-to-end architecture ensures accurate, interpretable, and clinically meaningful assessment of liver-cirrhosis progression.

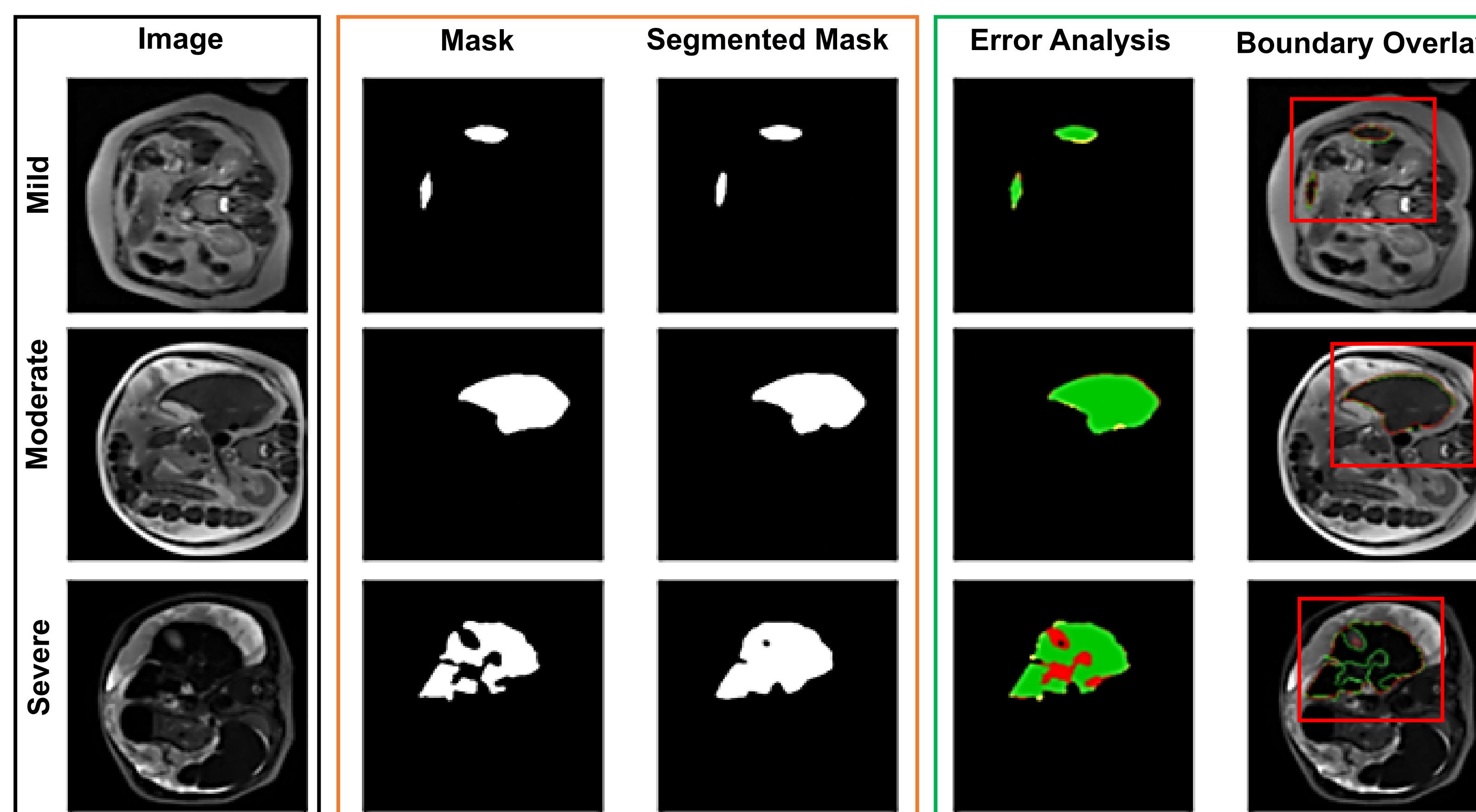


Figure 4: Comparison of the ground truth mask and predicted mask along with error analysis for the 3UResNet model.

The green regions indicate correctly detected cirrhotic areas, white denotes unlabeled or missed regions, and red highlights falsely segmented areas. Each row shows the MRI image, its mask, and boundary overlay for visual comparison of segmentation accuracy.

Results

Table 1: Performance evaluation of the proposed encoder-decoder architectures for liver-cirrhosis segmentation across mild, moderate, and severe stages (Dice and mIoU metrics).

Model		Mild		Moderate		Severe	
Encoder	Decoder	Dice	mIoU	Dice	mIoU	Dice	mIoU
ResNet50	UNet	0.9556	0.9150	0.9534	0.9109	0.9330	0.8745
	TransUNetPP	0.9532	0.9106	0.9578	0.9191	0.9310	0.8710
	Attention U-Net	0.9605	0.9239	0.9572	0.9180	0.9326	0.8738
	LinTransUNet	0.9416	0.8896	0.9559	0.9156	0.9282	0.8661
VGG16	U ² -Net	0.9315	0.8718	0.9411	0.8888	0.8563	0.7487
InceptionV3	SynergyNet	0.9575	0.9184	0.9546	0.9131	0.9339	0.8759
Swin CCT EaNet	UNet	0.9319	0.8725	0.9409	0.8884	0.9173	0.8472
		0.9321	0.8728	0.9347	0.8774	0.9281	0.8658
		0.9287	0.8669	0.9390	0.8850	0.9242	0.8592

Table 2: Performance evaluation of the proposed classification models for liver-cirrhosis severity prediction using visual and shape features.

Model Architecture			Mild				Moderate				Severe			
Feature Extractor		Classifier	Prec.	Rec.	F1-Score	Acc.	Prec.	Rec.	F1-Score	Acc.	Prec.	Rec.	F1-Score	Acc.
DenseNet-121	Regionprops	FFC	0.83	0.95	0.89	0.95	0.13	0.05	0.07	0.05	0.64	0.77	0.70	0.77
ConvNeXt	table		0.82	0.95	0.88	0.95	0.20	0.11	0.14	0.11	0.63	0.66	0.65	0.63
EfficientNet-B0			0.82	0.93	0.87	0.93	0.19	0.10	0.13	0.10	0.64	0.70	0.67	0.70
ResNet34			0.85	0.89	0.87	0.89	0.13	0.11	0.12	0.11	0.56	0.60	0.58	0.60
CoAtNet			0.78	0.86	0.82	0.86	0.21	0.14	0.17	0.14	0.51	0.54	0.52	0.54
ResNet50+GAP			0.81	0.88	0.84	0.88	0.19	0.13	0.15	0.13	0.66	0.72	0.69	0.72
WSP			0.76	0.95	0.85	0.9486	0.17	0.02	0.04	0.0226	0.64	0.75	0.69	0.7459
Hybrid3Net			0.91	0.71	0.80	0.7143	0.30	0.53	0.39	0.5338	0.59	0.50	0.54	0.5028
Hybrid6Net			0.90	0.71	0.80	0.7143	0.35	0.56	0.43	0.5639	0.71	0.69	0.70	0.6851
Hybrid7Net1			0.89	0.75	0.82	0.7514	0.55	0.49	0.52	0.4887	0.68	0.94	0.79	0.9448
Hybrid7Net2			0.86	0.93	0.90	0.9314	0.94	0.11	0.20	0.1128	0.60	0.90	0.72	0.9006
Hybrid7Net3			0.85	0.81	0.83	0.8057	0.68	0.60	0.64	0.6015	0.70	0.83	0.76	0.8287
Hybrid7Net4			0.94	0.73	0.82	0.7314	0.44	0.80	0.57	0.8045	0.78	0.64	0.70	0.6354
Hybrid7Net5			0.92	0.74	0.82	0.7371	0.47	0.56	0.52	0.5639	0.71	0.88	0.79	0.884
Hybrid7Net6			0.97	0.77	0.86	0.77	0.56	0.77	0.65	0.77	0.67	0.75	0.70	0.75
Hybrid7Net7			0.89	0.71	0.79	0.71	0.32	0.56	0.41	0.56	0.61	0.51	0.56	0.51
Hybrid10Net			0.84	0.91	0.88	0.9114	0.79	0.26	0.39	0.2556	0.58	0.77	0.66	0.7735

3UResNet achieved a Dice score of 0.9516 and a mIoU of 0.9077 for the segmentation task. For the classification task, **DenseNet121** performed best for mild cirrhosis with 95% accuracy, while **CoAnNet** achieved the highest accuracy for both moderate (80%) and severe (94%) cirrhosis compared to other evaluated approaches.

AI Explainability (XAI)

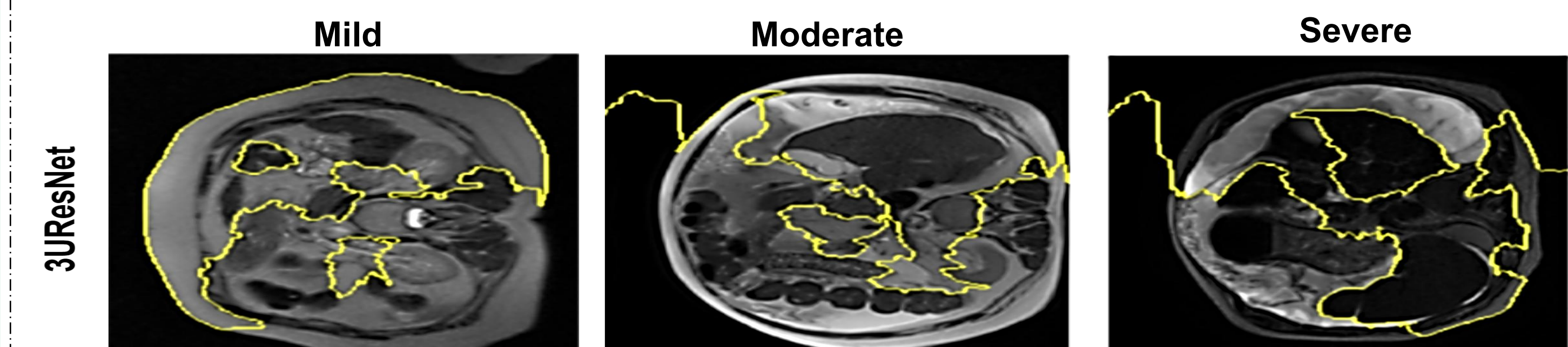


Figure 5: Segmentation efficiency in identifying key regions of the liver using LIME for explainability.

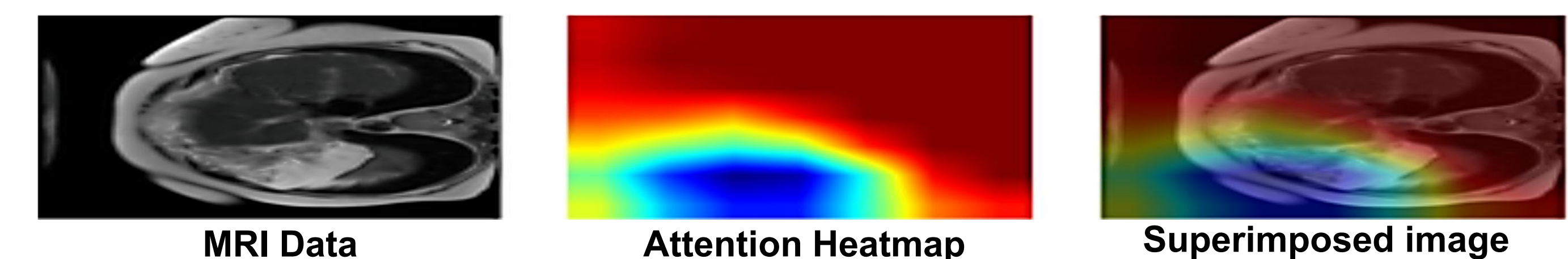
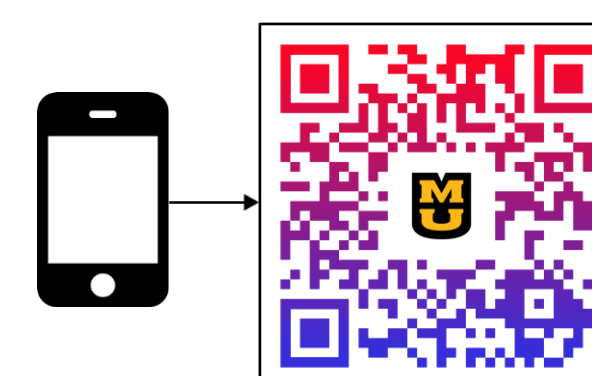


Figure 6: Grad-Cam Explainability for the Segmentation using 3UResNet.

Take a picture to download the code, data and poster



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